



Antonio Vasconcellos Mac Dowell

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in Antonio Mac Dowell 🌐 Website

Summary

I am a Mathematics PhD student pursuing a career focused on environmental causes. Experienced in developing numerical algorithms, risk management tools, scientific research and teaching.

Previous Work Experience

Instructor *PUC-Rio* [🔗](#). March 2024 - Dec 2024

- Discipline: Modeling and Programming (CTC4002) — Undergraduate course, 2 classes of 40 hours each.
- Description: Pre-calculus with an introduction to algorithmic thinking and Python programming.

Risk Analyst Intern *RJ+ Asset* [🔗](#). 2023 (summer)

- Member of a team developing algorithms for risk management, automating the computation of Value at Risk (VaR) for portfolio optimization using Python and Excel.

Software Engineer Intern Jan 2021 – Aug 2022

Fundação COPPETEC, Federal University of Rio de Janeiro [🔗](#).

- Member of an AI project for the State Public Defender's Office, Rio de Janeiro.
- Management of datasets, development of algorithms and meetings with clients.
- The project made significant contributions to the work of public defenders in Rio de Janeiro.

Education

PhD in mathematics *PUC-Rio* [🔗](#). 2024 – Present

- **Scholarship:** *Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES)* [🔗](#).
- **Research topic:** Integrable systems, Poisson geometry.
- **Seminars:** *Introduction to Toda lattices* [🔗](#) at *IMPA* [🔗](#).

Master's degree in Mathematics *PUC-Rio* [🔗](#). 2021 – 2024

- **Scholarship:** *CAPES* [🔗](#).
- **Dissertation:** *Recovery of Jacobi matrices from spectral data* [🔗](#).
- **Brief description:** Creation and development of an algorithm for inverse spectral theory of tridiagonal symmetric matrices; Numerical experiments showed superior performance compared to existing techniques.
- **Seminars:** *Random Matrices, Graphs and a combinatorial proof of Wigner's Semi-Circle Law* [🔗](#).

Bachelor's degree in Mathematics *PUC-Rio* [🔗](#). 2018 – 2021

- **Teaching Assistant** Calculus, Linear Algebra and Differential Equations.
- **Extra activities:** XII Competencia Iberoamericana Interuniversitaria de Matemáticas in 2020.

Skills

- **Languages:** English (Fluent), Portuguese (Native), French (Level B1 - Intermediate).
- **Informatics:** Python, SQL, HTML, CSS, LaTeX, Sage, Maple, MatLab, Excel, GitHub, etc.
- **Soft Skills:** Passionate and curious, good at speaking in public, Teamwork and explaining complex subjects.